

Phase : project testing phase

Date	29 september 2026
Team ID	SWTID-2026-9366
Project name	Script-controlled ACL
Maximum marks	4

The Project Testing Phase (or Quality Assurance Phase) validates that the developed system operates strictly according to business requirements, technical specifications, and quality standards before going live.

1. KeyObjectives

- Validate Requirements: Verify that all business rules, user stories, and security controls function as expected.
- Identify & Fix Defect: Uncover bugs, security vulnerabilities, and performance bottlenecks early to prevent production failures.
- Ensure System Stability: Confirm that individual modules integrate seamlessly and scale under expected workloads.
- Gain Stakeholder Approval: Demonstrate readiness to key business users through User Acceptance Testing (UAT).

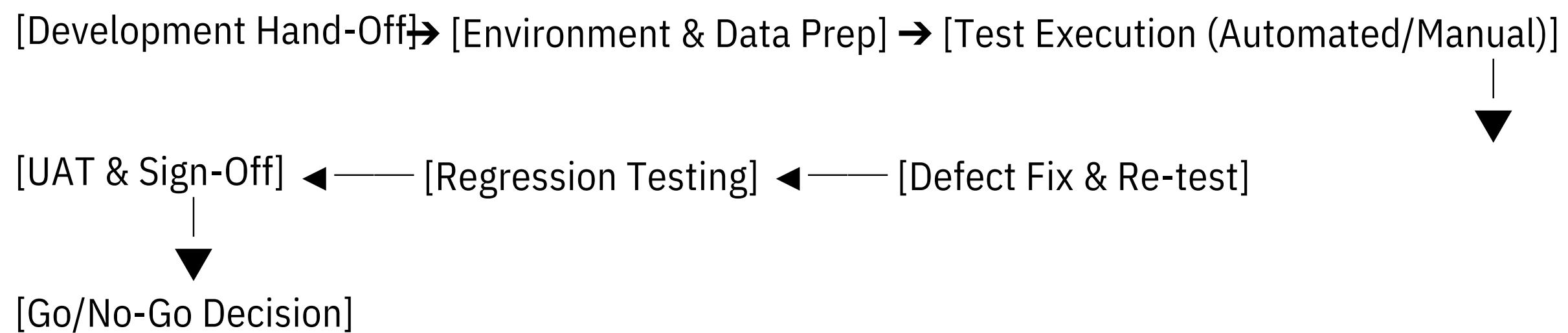
2. Core Activities & Deliverables

Category	Key Activities	Output Deliverables
Test Planning	Strategy definition, resource allocation, environment setup, test data generation.	Master Test Plan, Test Strategy Document
Test Case Authoring	Writing functional, integration, edge-case, and security test scenarios.	Test Scripts, Requirements Traceability Matrix (RTM)
Test Execution	Running manual and automated test suites, logging defects, regression testing.	Test Execution Logs, Defect Logs / Bug Reports
Performance & Security	Load testing, stress testing, vulnerability scanning, role-based access audits.	Security Audit Report, PerformanceBenchmarks
Validation & Sign-Off	UAT execution with business users, defect triaging, Go/No-Go evaluation.	UAT Sign-Off, Quality Summary Report

3. Core Testing Levels & Types

- Unit & Integration Testing: Validates individual code units and data flow between connected modules.
- System / Functional Testing: Evaluates end-to-end functionality against functional requirements.
- Security & ACL Testing: Validates Role-Based Access Control (RBAC), data isolation, and permission boundaries.
- Performance & Load Testing: Measures system latency, throughput, and stability under load spikes.
- User Acceptance Testing (UAT): End-users test actual business workflows in a production-like staging environment.

4. High-Level Process Workflow



1. Input: Stable Build, Deployment Packages, and Technical Design Specs.
2. Execution: Run smoke tests followed by deep functional, security, and integration test runs.
3. Defect Triage: Log issues into defect trackers (e.g., Jira), assign severity levels, and assign to developers for resolution.
4. Re-Testing & Regression: Verify fixed bugs and re-run main test flows to confirm fixes didn't introduce new issues.
5. Output: Fully validated software build, formal UAT sign-off, and deployment readiness approval.

5. Key Best Practices

- Traceability: Maintain a Requirements Traceability Matrix (RTM) mapping every test case back to its specific requirement.
- Automate Early: Automate repetitive regression and API test suites to run on continuous integration pipelines.
- Realistic Test Data: Utilize sanitized production-like data sets to capture realistic edge cases and performance impacts.

Are you looking to design test cases for a specific framework, automate continuous testing pipelines, or structure a UAT plan?